SENIOR AI ENGINEERING PREP

50+ Advanced RAG Interview Questions

The Ultimate Architecture & System Design Checklist to Clear Senior AI Interviews

7 Core Modules | 54 Technical Qs | Staff-Level Rigor

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SECTION 1: ADVANCED RETRIEVAL (1/3)

Retrieval Triggers & Correction

QUESTION 1

What is **Self-RAG** and how does it differ from standard RAG in terms of **retrieval triggers** and **output validation**?

QUESTION 2

What is **Corrective RAG (CRAG)** and what **three retrieval actions** does it decide between after scoring retrieved documents?

QUESTION 3

How does **FLARE** (Forward-Looking Active REtrieval) use **token probability** to trigger retrieval mid-generation?

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SECTION 1: ADVANCED RETRIEVAL (2/3)

Architectures & Hierarchies

QUESTION 4

What is **Agentic RAG** and how does it differ architecturally from a fixed pipeline RAG system?

QUESTION 5

Explain **Parent Document Retrieval** — why do you index small chunks but return large parent chunks to the LLM?

QUESTION 6

What is **RAPTOR**, how does it build its hierarchical tree during ingestion, and why does it outperform flat retrieval for multi-level questions?

QUESTION 7

How does **multi-vector retrieval** (e.g., ColBERT late interaction) differ from single-vector bi-encoder retrieval, and when is the extra cost justified?

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SECTION 1: ADVANCED RETRIEVAL (3/3)

Fusion, Compression & Routing

QUESTION 8

What is **ensemble retrieval**, what merging strategies exist (RRF, weighted, cascade), and when do you choose it over a single retriever?

QUESTION 9

What problem does **contextual compression** solve after retrieval, and how do you implement it without introducing too much latency?

QUESTION 10

What is **Adaptive RAG** and how does query classification drive routing to different retrieval strategies and k values?

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SECTION 2: CHUNKING STRATEGIES (1/3)

Semantic & Contextual Splits

QUESTION 11

What is **proposition-level chunking**, how does it differ from sentence or paragraph chunking, and what retrieval quality gains does it provide?

QUESTION 12

What is **"late chunking"** (as introduced by Jina AI), how does it differ from chunking before embedding, and what context information does it preserve?

QUESTION 13

What is Anthropic's **Contextual Retrieval** technique, and how does adding a short context prefix to each chunk before embedding improve recall?

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SECTION 2: CHUNKING STRATEGIES (2/3)

Code, Hierarchies & Windows

QUESTION 14

How do you chunk **source code files** differently from natural language documents — what boundaries do you split on and what metadata do you attach?

QUESTION 15

When chunking a document with multiple hierarchical levels (chapter → section → subsection → paragraph), what is the correct strategy to enable both coarse and fine retrieval?

QUESTION 16

What is the **sliding window chunking** approach, what overlap percentage range is typical, and what failure modes arise from too much versus too little overlap?

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SECTION 2: CHUNKING STRATEGIES (3/3)

Structured Data & Multi-Granularity

QUESTION 17

How would you chunk **structured data** like JSON blobs, XML records, or CSV rows for RAG — what representation makes them embeddable and retrievable?

QUESTION 18

What is **multi-granularity indexing** and how does storing the same content at different granularities simultaneously improve retrieval across different query types?

Pro Interview Tip

Explain how converting CSV rows into natural language key-value sentences dramatically boosts dense retrieval accuracy.

Core Principle

Never sever structured tags mid-record; keep JSON boundaries intact inside chunk metadata.

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SECTION 3: MULTIMODAL RAG (1/3)

Mixed Corpus & Page Vision

QUESTION 19

How would you design a RAG pipeline that retrieves relevant content from a corpus of documents containing **mixed text, images, tables, and charts**?

QUESTION 20

What is **ColPali**, how does it use a vision-language model to embed document page images directly, and what advantage does this have over text-extraction-based approaches?

QUESTION 21

What are the two main architectural approaches to multimodal document retrieval — **extract-then-embed vs. embed-the-image** — and what are the trade-offs of each?

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SECTION 3: MULTIMODAL RAG (2/3)

Unified Spaces & Video RAG

QUESTION 22

How do you generate embeddings for **images and text in the same vector space** so they can be retrieved together in a unified index?

QUESTION 23

How would you design a RAG system for **video content** — what are the chunking, embedding, and retrieval challenges specific to video?

QUESTION 24

What **evaluation metrics** are specific to multimodal RAG systems that go beyond text-only faithfulness and recall measures?

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SECTION 3: MULTIMODAL RAG (3/3)

Diagrams & VLM Roles

QUESTION 25

How do you handle **technical diagrams, flowcharts, circuit schematics, or architecture diagrams** in a RAG pipeline where visual content carries meaning not present in text?

QUESTION 26

What is the role of **vision-language models (VLMs)** like GPT-4V, LLaVA, or PaliGemma in document understanding for RAG, and when do you use them at ingestion vs. query time?

Ingestion Strategy

Use VLMs during batch ingestion to generate dense textual summaries of charts and schematics.

Query Strategy

Pass raw retrieved image crops directly to the multimodal LLM at query time for exact visual groundings.

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SECTION 4: GRAPHRAG & KGS (1/3)

Graph vs Vector & Core Architecture

QUESTION 27

What is **GraphRAG**, how does it fundamentally differ from vector RAG in how it represents and retrieves knowledge?

QUESTION 28

How does Microsoft's **GraphRAG** work at a high level — what does it build during indexing and how does it use community summaries for global questions?

QUESTION 29

How do you extract **entities and relationships** from raw text documents to build a knowledge graph for GraphRAG?

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SECTION 4: GRAPHRAG & KGS (2/3)

Community Summaries & Hybrid Routing

QUESTION 30

What are **community summaries** in GraphRAG, how are they generated, and what class of questions do they answer that flat vector retrieval cannot?

QUESTION 31

When should you choose GraphRAG over standard vector RAG — what **query types and corpus characteristics** make GraphRAG the right tool?

QUESTION 32

How do you combine **knowledge graph traversal with vector similarity search** in a hybrid retrieval system — what does query routing logic look like?

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SECTION 4: GRAPHRAG & KGS (3/3)

Scale, Indexing Cost & Evaluation

QUESTION 33

What are the main **indexing cost and storage challenges** of GraphRAG compared to vector RAG, and how do you mitigate them at scale?

QUESTION 34

How do you evaluate the quality of a GraphRAG system — what metrics capture **relationship retrieval accuracy** that standard Recall@k does not?

Indexing Cost Alert

Extracting entity-relationship triples across millions of chunks requires extensive LLM token burn during ingestion.

Global vs Local

GraphRAG shines on holistic dataset queries ("What are top themes?") where vector search retrieves scattered fragments.

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SECTION 5: FINE-TUNING VS RAG (1/2)

Decision Criteria & Hybrid Systems

QUESTION 35

What are the key decision criteria for choosing **fine-tuning over RAG**, and when is neither sufficient on its own?

QUESTION 36

How do you combine RAG with a fine-tuned model — what does fine-tuning add on top of retrieval, and what are the risks of this approach?

QUESTION 37

Compare the **cost, latency, and maintenance burden** of fine-tuning versus RAG across the entire model lifecycle.

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SECTION 5: FINE-TUNING VS RAG (2/2)

Component Tuning & Human Feedback

QUESTION 38

When building a domain-specific assistant, how do you decide whether to fine-tune the retriever, fine-tune the generator, implement RAG, or use all three?

QUESTION 39

What is the difference between fine-tuning an **embedding model vs. fine-tuning a generation model** in a RAG context, and when does each provide more value?

QUESTION 40

How does **RLHF or RLAIF** interact with a RAG system — can you use human feedback to improve a RAG pipeline without fine-tuning the underlying LLM?

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SECTION 6: MULTILINGUAL RAG (1/2)

Multilingual Models & Cross-Lingual Search

QUESTION 41

How do you design a RAG system that must handle documents and queries in **10+ languages** — what architectural decisions change compared to a monolingual system?

QUESTION 42

What are the best **multilingual embedding models** for RAG (e.g., multilingual-e5, LaBSE, mE5, BGE-M3) and what trade-offs do they make compared to English models?

QUESTION 43

What is **cross-lingual retrieval** — querying in English against French documents — and what strategies (shared embedding space, translation layer, separate indexes) handle it best?

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SECTION 6: MULTILINGUAL RAG (2/2)

Translation Approaches & Script Diversity

QUESTION 44

Compare the **translate-then-retrieve** and **retrieve-then-translate** approaches — what are the latency, quality, and cost implications of each?

QUESTION 45

How do you evaluate multilingual RAG quality across languages when your annotation team does not speak all languages in the corpus?

QUESTION 46

What chunking and preprocessing challenges arise specifically in multilingual corpora — script diversity, tokenization differences, right-to-left languages, and mixed-language documents?

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SECTION 7: AGENTIC RAG PATTERNS (1/2)

ReAct Loops & Tool Registries

QUESTION 47

What is the **ReAct (Reason + Act) pattern** and how does it structure the interaction between an LLM's reasoning steps and retrieval tool calls?

QUESTION 48

How do you build a RAG agent that can call **retrieval, a calculator, an API, and a code executor** in the same session — what is the tool registry design?

ReAct Flow

Thought → Action (Tool Call) → Observation → Thought → Final Answer.

Tool Routing

Equip the agent with distinct function schemas and clear docstrings to prevent hallucinated tool inputs.

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SECTION 7: AGENTIC RAG PATTERNS (2/2)

Iterative Retrieval & Multi-Agent Teams

QUESTION 49

What is **iterative retrieval** in agentic RAG — how does the agent decide to issue a second retrieval after evaluating the first result?

QUESTION 50

How do you design a **multi-agent RAG system** where domain-specialist agents (legal, financial, medical) handle their own retrieval and a coordinator merges answers?

Self-Correction

If retrieved context score is below threshold, agent reformulates query and re-searches automatically.

Coordinator Pattern

Supervisor LLM delegates sub-queries to isolated specialist agents, preventing vocabulary cross-contamination.

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THE NEXT STEP

You Have The Questions. Now Get The Answers.

Every single question in this carousel is broken down with step-by-step system diagrams, trade-off matrices, and production code in the kit:

- Detailed Architectural Answers for all 50+ Qs
- Working Production Code Examples & Prompts
- Complete System Design Case Studies

Check caption right now for full AI Interview Kit link